

Test 29: Dirac dip in DOS at $\eta=1/2$

Verdict: PASS | Generated: 2026-09-02 18:44:11 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

Semi-exact: DOS dip depth vs prediction.

METHOD: Semi-exact: DOS dip depth vs prediction.

Test 29 SERIES: 9-pt fine scan @96 η (strict dip true, $\eta_{\text{dip}}=0.0195$), contrast(w) 288.00 $\pm$ 3.97 ($w=0.1\pm0.8$), size contrast 13.33 $\times$ 10.67 $\times$

Result: [SERIES pass 3]: 3 sub-checks, 0 failed η PASS

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/plot_02' (.png 600dpi / .svg / .pdf)

- 'plots/plot_03' (.png 600dpi / .svg / .pdf)

- 'plots/plot_04' (.png 600dpi / .svg / .pdf)

- 'plots/animation.gif' η animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 29 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[17:57:51.474] [+14986.757s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\eta=0.3958$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:02:45.467] [+15280.749s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\eta=0.4271$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:07:45.953] [+15574.235s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\eta=0.4583$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:12:40.826] [+15876.108s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\eta=0.4896$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:17:38.995] [+16174.277s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\eta=0.5000$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:22:37.999] [+16473.281s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\eta=0.5104$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:27:30.820] [+16766.102s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\eta=0.5417$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:33:06.073] [+17101.353s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\eta=0.5729$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:38:26.831] [+17422.113s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\eta=0.6042$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:43:57.279] [+17752.561s] build_ab_cloud_hamiltonian: 48x48 (N=2304), $\eta=0.5000$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:44:03.436] [+17758.718s] build_ab_cloud_hamiltonian: 48x48 (N=2304), $\eta=0.3958$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

CONSOLE CAPTURE

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TEST 29 (pass 3, SERIES): Dirac dip ? fine ?-scan + window series

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Fine scan: 9 points ? = k/96 around k=48 (?=1/2 EXACT), window |E|<0.5

? RUNTIME: 9 solves at 96? + 2 anchors at 48? ? 8?13 min

?=38/96 = 0.39583: ?(|E|<0.5) = 0.2083 (289.4 s)

?=41/96 = 0.42708: ?(|E|<0.5) = 0.1458 (297.0 s)

?=44/96 = 0.45833: ?(|E|<0.5) = 0.0833 (290.1 s)

?=47/96 = 0.48958: ?(|E|<0.5) = 0.0208 (293.5 s)

?=48/96 = 0.50000: ?(|E|<0.5) = 0.0195 (294.9 s)

?=49/96 = 0.51042: ?(|E|<0.5) = 0.0208 (288.2 s)

?=52/96 = 0.54167: ?(|E|<0.5) = 0.0833 (331.0 s)

?=55/96 = 0.57292: ?(|E|<0.5) = 0.1458 (316.0 s)

?=58/96 = 0.60417: ?(|E|<0.5) = 0.2083 (326.3 s)

window |E|<0.1: ?_dip(?=1/2) = 0.0004, ?_off(?=38/96) = 0.1250 ? contrast 288.00×

window |E|<0.2: ?_dip(?=1/2) = 0.0039, ?_off(?=38/96) = 0.2083 ? contrast 53.33×

window |E|<0.3: ?_dip(?=1/2) = 0.0091, ?_off(?=38/96) = 0.2083 ? contrast 22.86×

window |E|<0.5: ?_dip(?=1/2) = 0.0195, ?_off(?=38/96) = 0.2083 ? contrast 10.67×

window |E|<0.8: ?_dip(?=1/2) = 0.0525, ?_off(?=38/96) = 0.2083 ? contrast 3.97×

contrast @w=0.5 (?=0.396 vs 1/2): 13.33×(48x48) ? 10.67×(96x96) ? holds/sharpens ?

??

strict dip @96? (9 pts) PASS ? ?(?=1/2) = 0.0195 below ALL 8 other scan points

contrast grows as w ? 0 PASS ? contrast(w) 288.00×(w=0.1) vs 10.67×(w=0.5): Dirac ? ? E vs metallic ? ? const

no wash-out 48??96? PASS ? 13.33×? 10.67×at the same ? pair (0.396 vs 1/2; tolerance 0.8, as 29b)

Test 29c [SERIES pass 3]: 3 sub-checks, 0 failed ? PASS